

1 PROBLEM / BACKGROUND

BEFORE: HARD TO USE



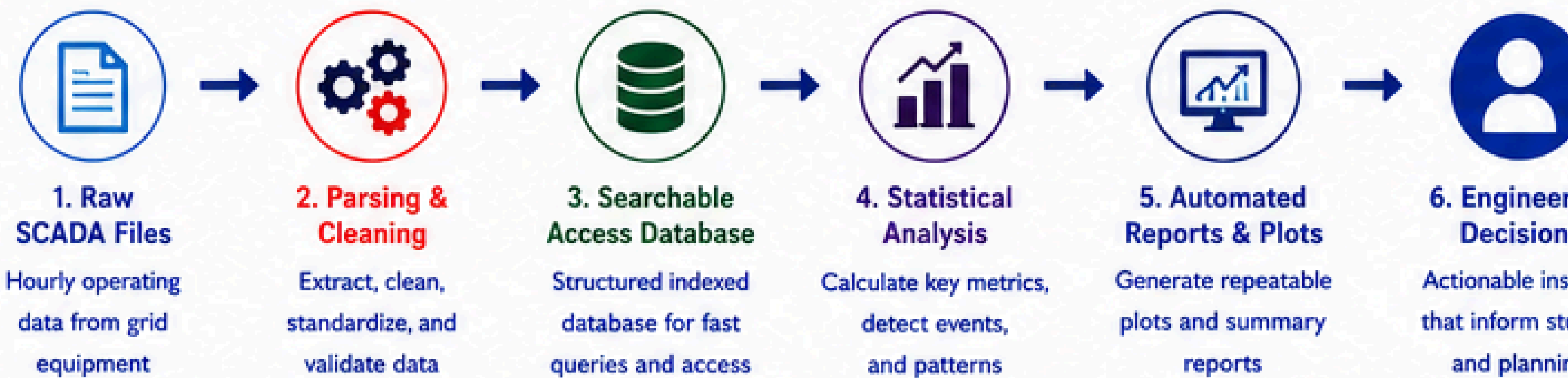
- Raw text files
- Hard to search or filter
- Time-consuming manual review
- Inconsistent formatting and naming
- Risk of missing data can distort results

AFTER: EASY TO USE



- Searchable database
- Clean, consistent data
- Automated analysis
- Clear visualizations
- Reliable insights for engineering decisions

2 WORKFLOW PIPELINE



An end-to-end pipeline converts raw historical grid data into clean, organized, and useful information engineers can quickly use.

3 ENGINEERING IMPACT

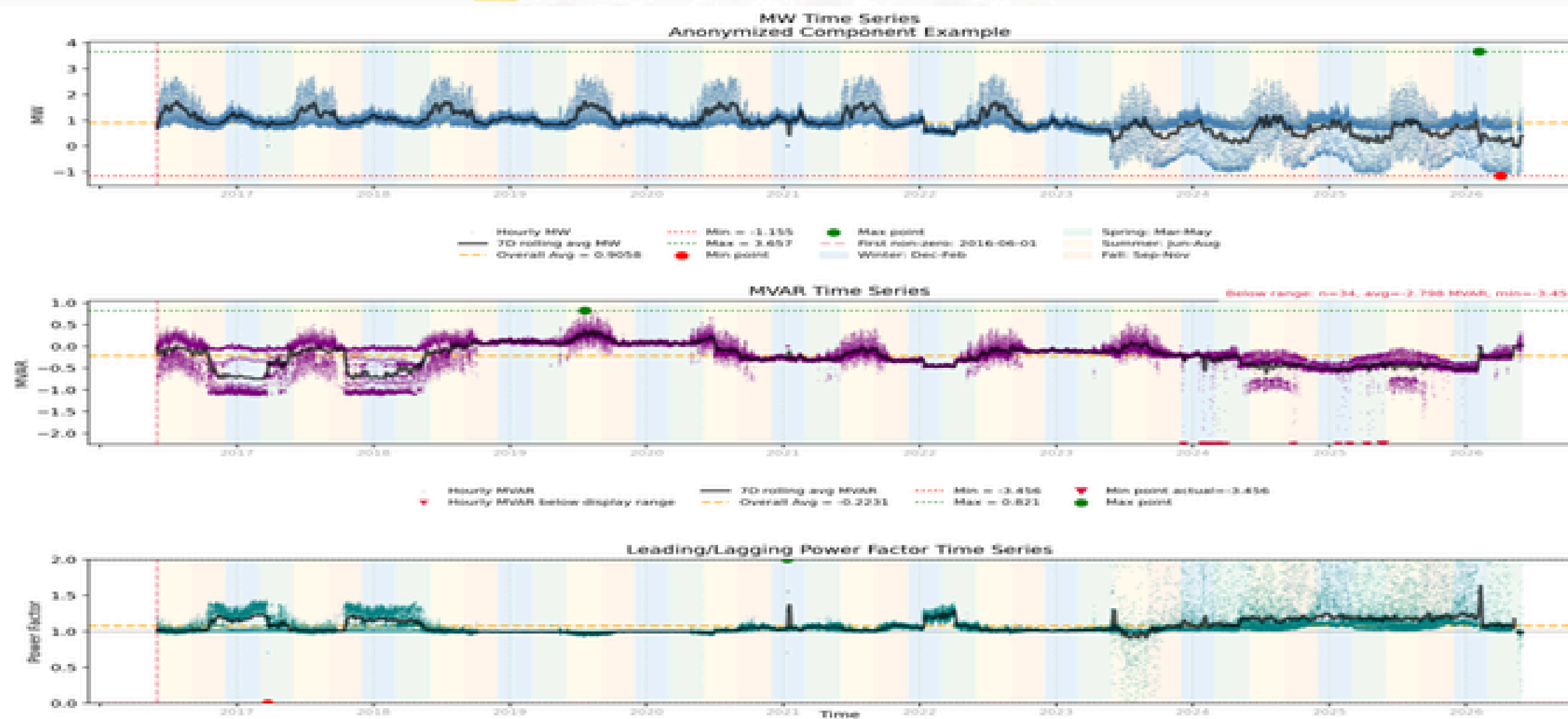
- Faster Studies**
Reduces time to pull seasonal and component-level data.
- Better System Models**
Supports more accurate and defensible models using real data.
- Reduced Planning Risk**
Improves confidence in grid behavior and planning decisions.
- Compliance Support**
Supports case building, load allocation, and UFLS-related analysis.

4 OBJECTIVE



Create a repeatable workflow that transforms 10 years of historical grid operating data into clean, searchable, and useful engineering insight for planning, modeling, and engineering studies.

5 RESULTS / EXAMPLE OUTPUT



n=87648 | MW std=0.5588, skew=-0.5291 | MVAR std=0.3172, skew=-1.044 | PF avg=1.073, std=0.141 | MW stats pts=87648, MVAR stats pts=87648

WHAT THE REPORTS REVEAL

- Seasonal behavior**
Recurring annual patterns are immediately visible.
- Unusual events**
Outliers and operating changes are automatically highlighted.
- Power-factor trends**
Long-term component behavior can be compared consistently.

6 KEY FEATURES



- Standardized component naming and metadata
- Automated quality checks and exception handling
- Searchable historical records for fast retrieval
- Reusable plotting and summary templates
- Consistent outputs for repeat engineering studies

8 CHALLENGES & LESSONS LEARNED



- Standardized data formatting across 10 years of data
- Filtered unusable telemetry values
- Handled missing, incorrect field labels; checks to reduce manual effort
- Met network and security constraints

9 FUTURE WORK



- Expand to additional data sources
- Add more real-time reporting and dashboards
- Integrate with analytics to flag engineering study insights

10 WHAT IS SCADA DATA?



SCADA data is hourly operating telemetry collected from grid devices across the electric grid. It includes values such as MW, MVAR, temperature, alarms, tap, names, and date-quality flags.

11 WORKFLOW BENEFITS



- Saves hours of MW, MVAR, and power factor slicing
- Identifies seasonal events and outliers
- Reduces rework and revisions
- Supports defensible engineering decision
- Scalable and future-expandable foundation



KEY TAKEAWAY: This project transformed 10 years of raw grid operating data into a searchable database and automated analysis workflow—helping engineers make **faster, more informed, and more confident decisions.**

